Ontology-Driven Semantic ETL for Executable Knowledge Engineering

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Abstract: Extract–Transform–Load (ETL) pipelines play a central role in integrating heterogeneous data sources for analytics and decision support. However, traditional ETL pipelines struggle to capture complex semantic relationships in unstructured data, while Large Language Models (LLMs), despite their extraction capabilities, lack logical rigor and are prone to hallucinations. This paper introduces an Ontology-Driven Semantic ETL, a hybrid semantic framework for transforming heterogeneous data into formal, executable knowledge models. The methodology separates informational structuring, semantic grounding, and knowledge materialization across the Extract, Transform, and Load layers, decoupling knowledge engineering into TBox induction and ABox population. It integrates LLM-based extraction with ontology alignment to produce logic-based models implemented in Prolog, enabling deterministic reasoning over domain knowledge. The approach is evaluated using real-world health-domain documents and compared with LLM-based knowledge graph and retrieval-based baselines. Results show that our approach supports complex deductive queries not expressible in conventional BI or vector-based systems, while improving precision and recall in entity and relation extraction. The resulting models provide greater completeness, consistency, and explainability through explicit semantic grounding. These findings highlight the importance of integrating ontology-based knowledge construction into ETL pipelines, bridging data engineering and knowledge representation, and enabling a verifiable alternative to standard RAG-based architectures.

1 Introduction

Extract–Transform–Load (ETL) pipelines are a cornerstone of modern data engineering, enabling the integration of heterogeneous data sources into structured repositories for analytics and decision support. Over time, ETL processes have evolved to incorporate increasingly sophisticated transformations, including the use of semantic technologies and, more recently, Large Language Models (LLMs) for extracting structured information from unstructured data.

Despite these advances, most ETL-based approaches remain primarily focused on data restructuring rather than knowledge construction. Even when semantic techniques are employed, the resulting representations are typically limited to enriched schemas or loosely structured data formats, which are insufficient to support explicit reasoning. Consequently, many data-driven systems, including business intelligence (BI) and AI applications, rely on retrieval-based mechanisms and contextual interpretation to derive insights from data.

This limitation becomes particularly evident in scenarios where meaning is implicit, distributed across multiple sources, and dependent on domain-specific relationships. In such settings, understanding how entities relate, interact, and constrain each other requires more than data integration; it requires the explicit representation of semantics. Traditional ETL pipelines, however, do not materialize these relationships during data preparation, leaving downstream systems responsible for inferring them at query time, often leading to variability, incompleteness, and limited explainability.

In this paper, we argue that ETL pipelines should go beyond data transformation and support the construction of formal knowledge representations. Rather than treating the Transform stage as a purely syntactic operation, we reinterpret it as a semantic construction process, in which domain concepts and relationships are explicitly modeled. This perspective shifts the role of ETL from data integration to knowl-

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edge engineering, enabling the generation of representations that can be directly used for reasoning.

To operationalize this idea, we propose an ontology-driven Semantic ETL approach that integrates LLM-based extraction with ontology alignment and logic-based modeling. The core contribution of this work is advancing Semantic ETL itself, incorporating ontology-based semantics into the pipeline, and enabling the transformation of heterogeneous data into formal and executable knowledge models. These models capture domain concepts and relationships in a structured and interpretable manner and are represented in a logic programming language, allowing systems to perform inference over the resulting knowledge base.

This approach has important implications for BI and AI systems. By materializing semantic relations during the ETL process, the proposed method enables analytical processes to operate over structured knowledge rather than raw structured data. This leads to more consistent, complete, and explainable outcomes, particularly in scenarios that require integrating information across multiple sources and interpreting implicit relationships.

The proposed methodology is evaluated from an ETL perspective, comparing its ability to construct semantically grounded knowledge representations with that of LLM-based and graph-oriented baselines. The results demonstrate that the ontology-driven approach produces representations that are more expressive, consistent, and suitable for reasoning, highlighting the benefits of integrating knowledge representation techniques into data engineering pipelines.

The main contributions of this paper are as follows: (i) an ontology-driven extension of Semantic ETL pipelines towards the construction of formal and executable knowledge models from heterogeneous data sources; (ii) an integration of LLM-based extraction, schema induction, and ontology alignment into a unified process that bridges data engineering and knowledge representation; and (iii) an evaluation demonstrating that the resulting representations enable logical inference in BI and AI systems, improving expressiveness, consistency, and interpretability when compared to existing approaches.

2 Background

Extract–Transform–Load (ETL) processes are a fundamental component of data engineering, supporting the integration of heterogeneous data sources into unified repositories for analytics and decision support. In this paradigm, the Extract phase retrieves data from multiple sources, the Transform phase applies cleaning and restructuring operations, and the Load phase materializes the transformed data into target systems (Kimball and Ross, 2013). Traditional ETL pipelines focus primarily on data integration and preparation through operations such as schema mapping, normalization, and aggregation (Inmon, 2005). Although modern ETL environments increasingly incorporate techniques for processing unstructured and semi-structured data, their primary objective remains data preparation rather than explicit knowledge construction.

Semantic ETL extends this paradigm by incorporating semantic technologies and ontologies to enrich data representations with meaning and improve interoperability (Lenzerini, 2002). Ontology-based data access systems demonstrate how semantic layers can bridge relational data and conceptual representations, allowing users to formulate queries at a higher level of abstraction (Calvanese et al., 2016). More recently, Large Language Models (LLMs) have been employed within ETL pipelines to extract entities and relationships from unstructured sources, supporting knowledge graph construction and semantic data integration (Liu et al., 2025). However, most Semantic ETL approaches remain focused on graph-based or schema-enriched representations, where semantics are only partially formalized and reasoning is typically deferred to query time.

Ontologies provide a formal specification of domain concepts and relationships, enabling the explicit representation of meaning in information systems (Guarino et al., 2009). A widely adopted definition describes an ontology as a formal, explicit specification of a shared conceptualization (Gruber, 1993), while ontological foundations have been used extensively in conceptual modeling to capture both structural and semantic aspects of a domain (Guizzardi, 2005). Formal frameworks such as Description Logics provide the theoretical basis for reasoning over ontological knowledge (Baader et al., 2003), whereas logic programming languages such as Prolog support executable knowledge representations through facts and rules (Kowalski and Smoliar, 1982). Knowledge bases are commonly divided into a terminological component (TBox) and an assertional component (ABox) (Baader et al., 2003). The TBox defines concepts, relationships, and constraints, whereas the ABox contains instance-level facts. This distinction enables the construction of semantically grounded and computationally actionable knowledge models capable of supporting explicit reasoning over heterogeneous information sources.

3 Related Work

Semantic ETL extends traditional data integration processes by incorporating semantic information into transformation pipelines. Early approaches focused on mapping relational data into RDF representations, formalized through standards such as R2RML (Consortium et al., 2012), enabling the construction of knowledge graphs from structured sources. Subsequent work explored semantic annotations and metadata enrichment to improve interoperability across heterogeneous systems. More recently, Large Language Models (LLMs) have been employed to automate semantic extraction from unstructured data, supporting knowledge graph construction and semantic data integration (Freire et al., 2025). Despite these advances, most Semantic ETL approaches remain centered on graph-based representations, where semantics are implicitly encoded in graph structures and reasoning is typically limited to traversal and pattern matching.

Ontologies have been widely adopted to provide semantic consistency and conceptual clarity in data integration environments. Ontology-based integration approaches use domain ontologies as conceptual mediators between heterogeneous data sources, enabling unified access and interpretation (Wache et al., 2001). Ontology matching and alignment techniques further support semantic interoperability by establishing correspondences between schemas and conceptual models (Euzenat and Shvaiko, 2007). In practice, however, ontologies are often employed primarily as annotation or mediation layers, rather than as foundations for executable knowledge representations capable of supporting reasoning and decision support.

The construction of executable knowledge models has been explored in knowledge representation, rule-based systems, and logic programming. Rule-based approaches demonstrated how declarative rules can encode domain logic and support automated reasoning (Forgy, 1989), while semantic web technologies introduced languages such as SWRL to combine ontologies with rule-based inference (Horrocks, 2004). More recent efforts have investigated the integration of knowledge graphs with logical reasoning frameworks, seeking to combine representational flexibility with formal semantics. Nevertheless, such approaches frequently require additional interpretation or transformation layers before executable reasoning can be performed. In contrast, logic programming environments such as Prolog provide a direct mechanism for representing executable knowledge through facts and rules evaluated by inference engines.

Existing work has significantly advanced semantic enrichment, ontology-based integration, and knowledge graph construction. However, a gap remains between semantic enrichment and executable knowledge construction. Most approaches generate semantically enriched representations but do not produce formal models that can be directly used for logical inference, consistency checking, or rule-based decision support. The approach proposed in this paper addresses this gap by integrating LLM-based extraction, schema induction, ontology alignment, and logic-based representations into a Semantic ETL pipeline that produces executable knowledge models, advancing Semantic ETL from data enrichment toward knowledge construction.

4 Methodology

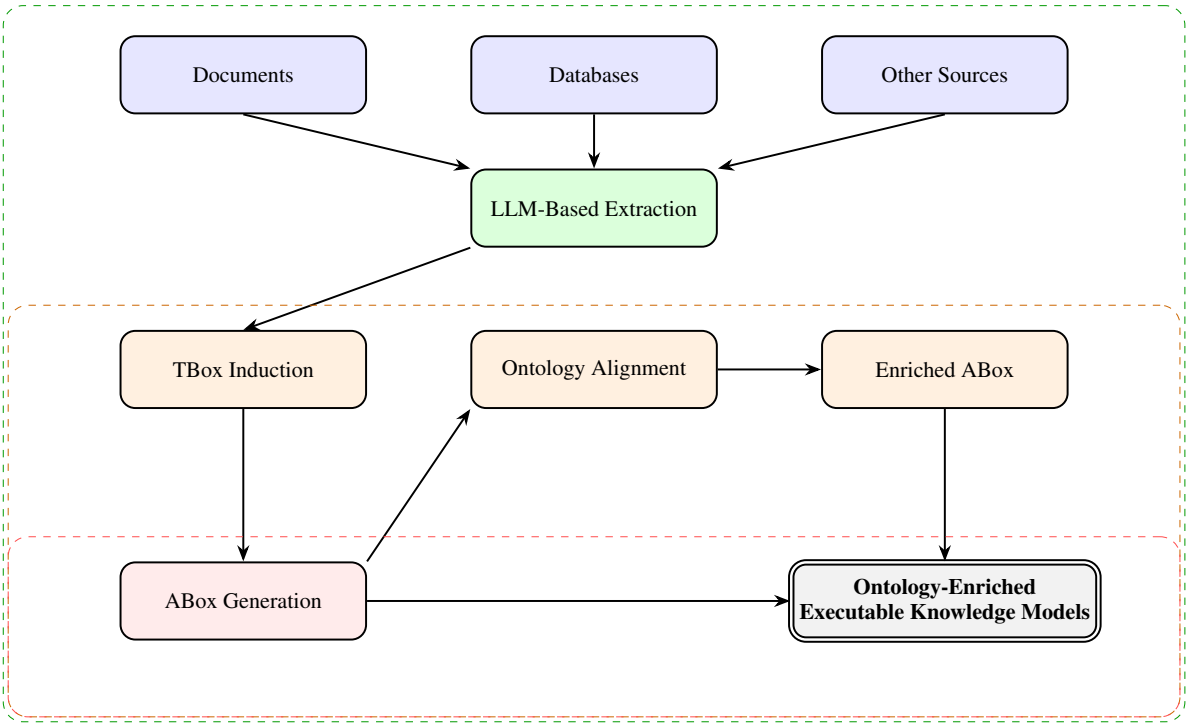
While Semantic ETL pipelines have been explored in prior work, they are typically focused on enriching data representations with semantic annotations or integrating heterogeneous sources at the schema level. In contrast, the methodology proposed in this paper focuses on constructing formal and executable knowledge models as the primary outcome of the ETL process. In this sense, the pipeline does not merely transfer data from source systems to a target repository; it progressively constructs a semantic substrate that can support reasoning, interpretation, and decision-making.

4.1 Ontology-Driven Semantic ETL Pipeline

Figure 1 presents the proposed Semantic ETL pipeline, organized into the three classical conceptual layers: *Extract*, *Transform*, and *Load*. The pipeline adapts the classical ETL paradigm to support ontology-based knowledge construction, extending traditional data engineering processes with semantic modeling and enrichment capabilities. Each layer produces explicit intermediate artifacts that serve as inputs to subsequent stages, enabling traceability and reproducibility of the knowledge construction process.

Extract Layer The Extract layer is responsible for acquiring and structuring information from heterogeneous data sources as a preparatory step toward the construction of formal, executable knowledge models. Unlike traditional ETL pipelines, which typically operate over structured data sources, the proposed approach explicitly targets heterogeneous inputs where

Figure 1: Semantic ETL Methodology for ontology-based knowledge construction from heterogeneous data sources.



relevant information is often embedded in natural language and semi-structured artifacts.

To address this challenge, the pipeline employs a Large Language Model (LLM)-based extraction component that transforms raw artifacts into structured representations by identifying candidate entities, attributes, and relationships. This step focuses on making implicit informational content explicit and consistently organized, without yet committing to any formal or ontological interpretation.

From a methodological perspective, this step can be understood as a structured information extraction procedure, where:

- the input consists of raw heterogeneous artifacts;
- the output consists of structured representations containing candidate entities, attributes, and relations;
- the extraction process is guided by prompt-based instructions that constrain the LLM to produce consistent and normalized outputs suitable for downstream processing.

The output of this stage is not a formal knowledge model, nor does it incorporate domain semantics in a principled way. Instead, it corresponds to a normalized intermediate representation that captures the informational content required for subsequent knowledge induction and semantic grounding. This intermediate layer establishes a stable bridge between heterogeneous data sources and the formalization processes that follow.

Transform Layer The Transform layer is responsible for constructing the semantic structure that will guide downstream knowledge representation. This layer includes two main components: *TBox induction* and *ontology alignment*, which together define the formal and semantic foundations of the target knowledge model.

TBox induction defines the schema of the target knowledge representation, specifying predicates, relations, and structural constraints. This process is performed by analyzing representative extracted samples and identifying recurring structural patterns. Methodologically, this step corresponds to schema induction, producing a formal specification of predicate signatures and permissible relations.

Ontology alignment complements this process by mapping the induced schema and extracted elements to concepts defined in a domain ontology. This step introduces explicit semantics into the pipeline, enabling the representation of roles, relationships, and domain-specific concepts such as obligations, rights, and actions. This alignment operates as a semantic transformation step that:

- maps syntactic structures to ontological concepts;

- resolves ambiguities in the extracted data;
- enriches representations with domain semantics not explicitly present in the source artifacts.

Importantly, while ontology alignment may operate over both schema-level elements and extracted instances, the actual materialization of instance-level knowledge (ABox) is not performed in this layer. Instead, the Transform layer defines the semantic and structural conditions under which such instances can be generated.

The output of the Transform layer is therefore a semantically grounded schema, together with alignment rules that enable the consistent transformation of extracted representations into formal instances. These specifications provide the necessary foundation for ABox generation, which is carried out in the subsequent Load layer.

Load Layer The Load layer is responsible for materializing instance-level knowledge according to the schema defined in the previous stage. In the proposed approach, this is achieved through *ABox generation*, which populates the knowledge base with contract-specific facts derived from the extracted content.

From a methodological standpoint, ABox generation can be understood as an *instance construction procedure*, where:

- the TBox provides structural constraints;
- the extracted representations provide content;
- the LLM generates structured facts that conform to the induced schema.

The resulting ABox corresponds to a structurally valid, but not yet fully semantically grounded, representation of the source artifacts. In particular, while the generated instances conform to the induced schema, they do not yet incorporate the full set of ontological commitments required for domain-level interpretation.

To address this, the generated ABox is subsequently subjected to ontology alignment procedures defined in the Transform layer. These procedures enrich the instance-level representation by introducing explicit domain semantics, including concepts such as obligations, rights, roles, and enforcement relations. Importantly, the Transform layer specifies how such semantic enrichment should be performed, while the Load layer provides the concrete instances over which these transformations operate.

The final output of this stage is an *ontology-enriched ABox*, which constitutes an executable knowledge model capable of supporting logical inference and semantically grounded queries.

5 Evaluation

The evaluation compares the proposed ontology-driven Semantic ETL pipeline with a representative baseline based on recent LLM-driven approaches for knowledge construction, particularly those focused on knowledge graph generation from heterogeneous data sources (Liu et al., 2025). While such approaches extend traditional ETL pipelines by introducing semantic structuring capabilities, they typically produce graph-based representations that remain limited in terms of formal semantics and reasoning support.

From an ETL perspective, both approaches can be interpreted as pipelines that progressively transform heterogeneous data into structured representations. However, they differ fundamentally in how semantic transformation is performed and materialized. In LLM-based approaches, semantic interpretation is largely implicit and embedded in graph structures, whereas in the proposed approach, semantic relations are explicitly constructed and grounded in an ontology, resulting in executable knowledge models. To operationalize the formal knowledge models produced by the Transform and Load layers, the proposed approach adopts Prolog as the underlying execution environment. In this setting, the induced TBox is implemented as a set of predicate definitions and structural constraints, while the generated ABox corresponds to a collection of ground facts instantiated from the extracted data. This representation enables the direct execution of logical queries over the resulting knowledge base, supporting inference mechanisms that go beyond simple data retrieval. In particular, Prolog allows the evaluation of complex queries involving variable binding, rule-based inference, and recursive reasoning, which are not natively supported in conventional data processing or BI systems. The use of Prolog thus reinforces the notion of executable knowledge models, where semantics are not only explicitly represented but also computationally actionable, enabling transparent and reproducible reasoning over domain-specific concepts and relationships.

The evaluation is therefore conducted by analyzing the evolution of artifacts across the ETL pipeline and assessing the extent to which each approach supports structured knowledge construction. As previously discussed, the Extract stage produces structured representations derived from heterogeneous sources using LLM-based extraction. The Transform stage introduces schema induction and ontology alignment, enabling the explicit representation of domain semantics. Finally, the Load stage materializes ontology-enriched ABoxes, forming executable knowledge models capable of supporting inference.

Table 1: Comparison between LLM-based Semantic ETL and ontology-driven Semantic ETL.

Dimension	**LLM-Based Semantic ETL (Baseline)**	**Proposed Approach**
Schema Representation	Implicit or graph-based structures	Explicit TBox with formally defined predicate signatures
Semantic Enrichment	Entity and relation extraction from text	Ontology-aligned semantic relations grounded in domain concepts
Knowledge Representation	Knowledge graphs (triples)	Logic-based executable knowledge models
Reasoning Capability	Limited to graph traversal and pattern matching	Logical inference over formally defined predicates
Transformation Objective	Structuring heterogeneous data	Constructing semantically grounded knowledge

In contrast, the baseline approach can be interpreted as a form of Semantic ETL in which LLMs are used to extract entities and relations and organize them into knowledge graphs. In this case, the Extract stage performs LLM-based information extraction, the Transform stage maps extracted elements into graph structures, and the Load stage materializes these structures as knowledge graphs. Although this process introduces a degree of semantic organization, the resulting representations remain largely structural and do not enforce explicit domain semantics or support formal reasoning.

The key differences between the two approaches are summarized in Table 1. The comparison highlights that, while both pipelines incorporate semantic elements, they diverge significantly in terms of representation, expressiveness, and reasoning capabilities. The baseline relies on graph-based representations, where semantics are implicitly encoded in node and edge labels, whereas the proposed approach produces logic-based models with explicitly defined predicates and ontological grounding.

A central aspect of the evaluation concerns the degree of semantic enrichment introduced during the Transform and Load stages. In the baseline approach, semantic information is implicitly captured through extracted relations, but remains dependent on textual interpretation and graph structure. In contrast, the proposed approach introduces explicit semantic grounding through ontology alignment, enabling the representation of domain concepts, constraints, and relationships in a formal and interpretable manner. This results in representations that are not only richer in meaning, but also more suitable for systematic processing. To further assess the practical capabilities of the proposed approach, we complement the analysis with three additional dimensions: (i) the ability to support complex queries beyond conventional BI systems, (ii) an error analysis of LLM-induced artifacts, and (iii) a quantitative evaluation of extraction performance using real-world unstructured sources.

Listing 1: Illustrative extract of the Prolog TBox.

```
% ---------------------------------------------
% TBox-style predicate signatures
% ---------------------------------------------
predicate_signature(disease, 1).
predicate_signature(caused_by, 2).
predicate_signature(common_symptom, 2).
predicate_signature(vaccine_protects_against, 2).
predicate_signature(may_cause, 2).
predicate_signature(most_people_recover_without, 2).
predicate_signature(severe_symptoms_require, 2).
predicate_signature(global_reported_cases, 2).
predicate_signature(global_reported_deaths, 2).
predicate_signature(statistic_reference_date, 2).
```

Complex Query Support. Listings 1 and 2 illustrate an excerpt of the executable Prolog knowledge model generated from the WHO COVID-19 fact sheet. The example shows how textual statements originally expressed in natural language are transformed into explicit logical structures, including schema-level predicate definitions and instance-level facts. This representation allows for the operationalization of information that would otherwise remain embedded in prose, such as causal relations, symptoms, transmission conditions, recovery expectations, vaccination effects, and epidemiological statistics. From an evaluation perspective, this example highlights an important advantage of the proposed approach: the resulting model is not merely a structured representation for storage, but an executable artifact that supports logical querying and inference. In contrast to conventional BI pipelines, where such content would require prior manual schema design and extensive preprocessing, the generated Prolog model enables direct inspection of disease characteristics and their relations in a formally grounded and computationally actionable way.

A key advantage of the proposed ontology-driven

Listing 2: Illustrative extract of the Prolog ABox.

```
% ------------------------------------------
% ABox facts extracted from the WHO fact sheet
% ------------------------------------------
disease(covid_19).
caused_by(covid_19, sars_cov_2).

common_symptom(covid_19, fever).
common_symptom(covid_19, chills).
common_symptom(covid_19, sore_throat).

spreads_between_people_in_close_contact(covid_19).

vaccine_protects_against(covid_19, severe_illness).
vaccine_protects_against(covid_19, death).

may_cause(covid_19, serious_illness).
may_cause(covid_19, death).

most_people_recover_without(covid_19, treatment).
severe_symptoms_require(covid_19,
medical_care_as_soon_as_possible).

global_reported_cases(covid_19, 760000000).
global_reported_deaths(covid_19, 6900000).
global_vaccine_doses_administered(covid_19,
13000000000).
```

pipeline lies in its ability to support complex, semantically grounded queries that go beyond the capabilities of traditional BI systems. Using unstructured reports from the World Health Organization (WHO) as input, such as COVID-19 (WHO, 2024a) and dengue (WHO, 2024b), the resulting executable knowledge models enable queries that require explicit reasoning over domain concepts and relationships. For example, the model supports queries such as: (i) identifying all diseases associated with viral transmission and their corresponding transmission vectors, (ii) inferring whether a disease involves human-to-human transmission or vector-borne mechanisms, and (iii) retrieving conditions under which a disease may evolve into a severe form based on explicitly modeled criteria. Such queries require the combination of multiple semantic relations and cannot be expressed through conventional SQL queries over structured tables without prior manual modeling. In contrast, the baseline approach, based on knowledge graph generation, allows traversal-based queries but lacks the formal semantics required to support inference over implicit domain constraints.

Error Analysis of TBox Induction and Ontology Alignment. Despite its advantages, the proposed pipeline is subject to limitations arising from the use of LLMs in the Extract and Transform stages. In particular, errors were observed in both TBox induction and ontology alignment when processing WHO textual data. In some cases, the LLM failed to correctly generalize schema-level patterns, producing inconsistent predicate definitions (e.g., mixing attributes and relations or generating overly specific predicates tied to individual documents). Additionally, ontology alignment occasionally introduced logical inconsistencies, such as misclassifying transmission mechanisms (e.g., incorrectly aligning vector-borne transmission with direct human transmission) or failing to distinguish between symptoms and disease outcomes. These errors highlight the sensitivity of the pipeline to prompt design and the need for validation mechanisms to ensure logical consistency. Importantly, however, such inconsistencies become detectable due to the explicit nature of the resulting knowledge models, enabling systematic inspection and correction, in contrast to purely implicit representations in LLM-based baselines.

Quantitative Evaluation of Extraction Performance. To evaluate the effectiveness of the extraction process, we conducted a comparative analysis of entity and relation extraction using WHO disease descriptions (e.g., COVID-19 and dengue reports) as ground truth references. Precision and recall were computed by comparing the extracted entities (e.g., disease names, symptoms, transmission modes) and relations (e.g., *caused_by*, *transmitted_by*, *has_symptom*) against manually curated annotations.

Table 2 summarizes the results. For comparison purposes, two lightweight extraction baselines were implemented. The rule-based baseline relies on manually defined domain lexicons and keyword-matching rules to identify entities and relations. The pattern-based baseline extends this approach through the use of regular-expression patterns and syntactic templates designed to capture common disease, symptom, and transmission expressions found in WHO reports. These baselines were selected because they represent conventional information extraction strategies frequently employed in ETL preprocessing pipelines and provide a reference point for assessing the benefits of LLM-based extraction. The proposed LLM-based extraction achieved high precision in entity identification, reflecting its ability to capture salient concepts from unstructured text. However, recall varied depending on the complexity of the textual descriptions, with some implicit relations not being fully captured. Compared to traditional rule-based or pattern-based extraction methods, the proposed approach demonstrated higher coverage of semantically relevant entities and relations, particularly in cases where information was distributed across multiple sentences or expressed implicitly.

Table 2: Extraction performance comparison on WHO disease descriptions.

Approach	Entities		Relations	
	Precision	Recall	Precision	Recall
Rule-based Extraction	0.78	0.61	0.72	0.54
Pattern-based Extraction	0.81	0.65	0.75	0.58
LLM-based Extraction (proposed)	**0.91**	**0.84**	**0.88**	**0.79**

These results indicate that LLM-based extraction provides a strong foundation for subsequent semantic grounding, particularly due to its ability to capture implicit and context-dependent information. Nevertheless, the observed gaps in recall for certain relation types suggest that additional mechanisms, such as iterative refinement or ontology-guided extraction, may be required to ensure completeness and consistency.

Together, these additional analyses reinforce the central claim of this work: while LLM-based approaches enable flexible extraction from heterogeneous sources, it is the integration with ontology-driven semantic grounding and formal knowledge representation that enables reliable, interpretable, and executable knowledge construction.

6 Conclusions and Future Work

This paper presented an ontology-driven approach to Semantic ETL for transforming heterogeneous data into formal and executable knowledge models. By integrating LLM-based extraction, schema induction, ontology alignment, and logic-based representations, the proposed pipeline extends traditional ETL from data preparation toward knowledge construction. The evaluation showed that, compared with LLM-based and graph-oriented baselines, the resulting models provide greater semantic grounding, interpretability, and reasoning capabilities, demonstrating the benefits of incorporating ontology-based semantics into data engineering workflows.

Future work will investigate schema evolution mechanisms for ontology-driven ETL pipelines and evaluate the proposed approach in truly heterogeneous environments involving documents, relational databases, and multimodal sources. Additional research will focus on quantitative metrics for assessing semantic coverage, integration quality, and knowledge completeness, as well as on hybrid reasoning approaches that combine symbolic inference with statistical and neural methods to further enhance Semantic ETL in BI and AI applications.

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